

number			
7(a)	An answer that includes two of the following points: - because it would be {unethical / wrong} to {kill / harm} the insects (1) - because the insects would contain {protein / amino acids} (1)		(2)

	it would give an {incorrect / higher} value (for the {protein / amino acid} content of the galls) (1)		
--	---	--	--

Question number	Answer	Mark
7(b)(i)	The only correct answer is B. <i>A is incorrect because aspartate is the most polar molecule.</i> <i>C is incorrect because aspartate is the most polar molecule.</i> <i>D is incorrect because aspartate is the most polar molecule.</i>	(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	A calculation in which: - solubility of leucine calculated (1) - solubility comparison with histidine calculated (1)	Example of calculation: 5.5g in 250 cm³ = 22.0 (g dm⁻³) 43.5 ÷ 22 = 1.98 / 2.0	(2)

number		
*7(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • there is a high concentration of protein in the galls • because the saliva stimulates protein synthesis • so there is a store of protein for the insect • there is a high concentration of amino acids in the galls • because the amino acids are needed for protein synthesis	(6)

		• because the enzymes in the saliva were breaking proteins down • for use by the insects • {alanine / arginine / histidine} are very abundant in the galls • because these amino acids are abundant in the protein in the galls • none of the amino acids are abundant in the tissues of the leaf with galls • because they have moved into the gall • leucine and tryptophan are not abundant in the galls • as they have been used by the insects	
Level	Marks		
	0	No awardable content.	
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	
2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning with some structure.	
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically	

		structured.
--	--	-------------